TryHackMe_Networking_Concepts.md

This module explores core networking protocols, addressing schemes, transport mechanisms (TCP vs. UDP), packet encapsulation across layers, and practical application testing using telnet.

Task 3: The TCP/IP Model (5-Layer Model)

Modern networking literature (such as Kurose & Ross) often models the Internet architecture using 5 layers:
Application Layer (HTTP, FTP, Telnet)
Transport Layer (TCP, UDP)
Network Layer (IP, ICMP)
Link Layer (Ethernet, Wi-Fi)
Physical Layer

Questions & Answers:

To which layer does HTTP belong in the TCP/IP model?
Answer: Application
How many layers of the OSI model does the application layer in the TCP/IP model cover?
Answer: 3 (Application, Presentation, and Session layers)

Task 4: IP Addresses and Subnets

IP addressing relies on reserved ranges for private internal networks (RFC 1918) and strict range limits for IPv4 octets (0–255).

Private IPv4 Ranges:
10.0.0.0 – 10.255.255.255
172.16.0.0 – 172.31.255.255
192.168.0.0 – 192.168.255.255

Questions & Answers:

Which of the following IP addresses is not a private IP address?
Options: 192.168.250.125, 10.20.141.132, 49.69.147.197, 172.23.182.251
Answer: 49.69.147.197
Which of the following IP addresses is not a valid IP address?
Options: 192.168.250.15, 192.168.254.17, 192.168.305.19, 192.168.199.13
Answer: 192.168.305.19 (305 exceeds the valid octet limit of 255)

Task 5: UDP and TCP

TCP provides reliable, connection-oriented data transfer through a mandatory 3-way handshake (SYN \rightarrow SYN-ACK \rightarrow ACK), utilizing port numbers from 1 to 65,535 (~65 thousand ports).

Questions & Answers:

Which protocol requires a three-way handshake?
Answer: TCP
What is the approximate number of port numbers (in thousands)?
Answer: 65

Task 6: Encapsulation

As data travels down the stack, each layer encapsulates the data unit with its own protocol header:

Application Data \rightarrow Datagram (UDP) or Segment (TCP) \rightarrow IP Packet \rightarrow Link Frame (Ethernet / Wi-Fi).

Questions & Answers:

On a WiFi, within what will an IP packet be encapsulated?
Answer: Frame
What do you call the UDP data unit that encapsulates the application data?
Answer: Datagram
What do you call the data unit that encapsulates the application data sent over TCP?
Answer: Segment

Task 7: Telnet

Telnet is an unencrypted text-based protocol used to establish interactive remote sessions or manually craft HTTP requests (e.g., GET / HTTP/1.1) to inspect server response headers.

Practical Workflow:

Connect to the web server using Telnet: telnet MACHINE_IP 80.
Issue a manual HTTP GET request:
GET / HTTP/1.1
Host: telnet.thm

Analyze the HTTP 200 OK response headers to identify the web server name/version and retrieve the target flag.

Questions & Answers:

Use telnet to connect to the web server on MACHINE_IP. What is the name and version...?
Answer: (Exacted server banner from response headers, e.g., Apache/2.4.41)
What flag did you get when you viewed the page?
Answer: THM{...} (Extracted flag string contained within the HTML body)